

WEARABLE SYSTEM FOR MONITORING POSTURE AND HUMAN MOVEMENTS

MS1 17/03/2026



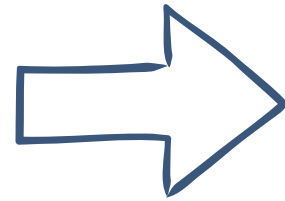
Beatriz Francisco N° 118638
Catarina Ribeiro N° 119467
Mariana Marques N°118971
Marta Cruz N° 119572
Matilde Rodrigues N°119714

MOTION SUIT

Our project involves developing a suit equipped with sensors that measure body posture and detect anomalies



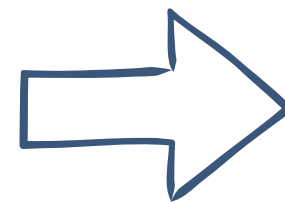
PERSONAS



As a
Suit User

I want
View statistics on
dashboard

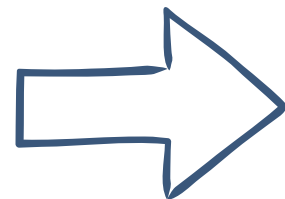
So that
I can see my posture data
over time



Researcher

Access all collected
information

I can identify postural
deviations and abnormal
patterns

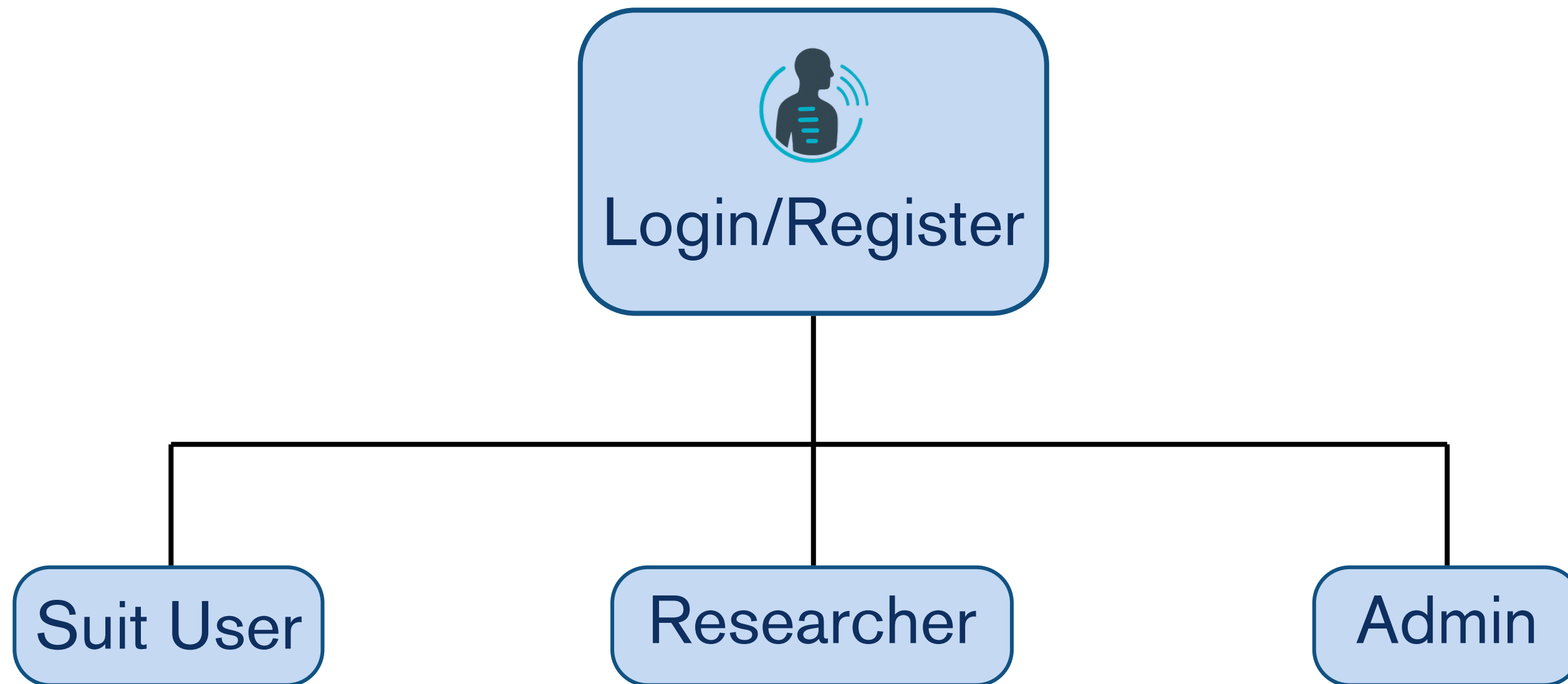


Administrator

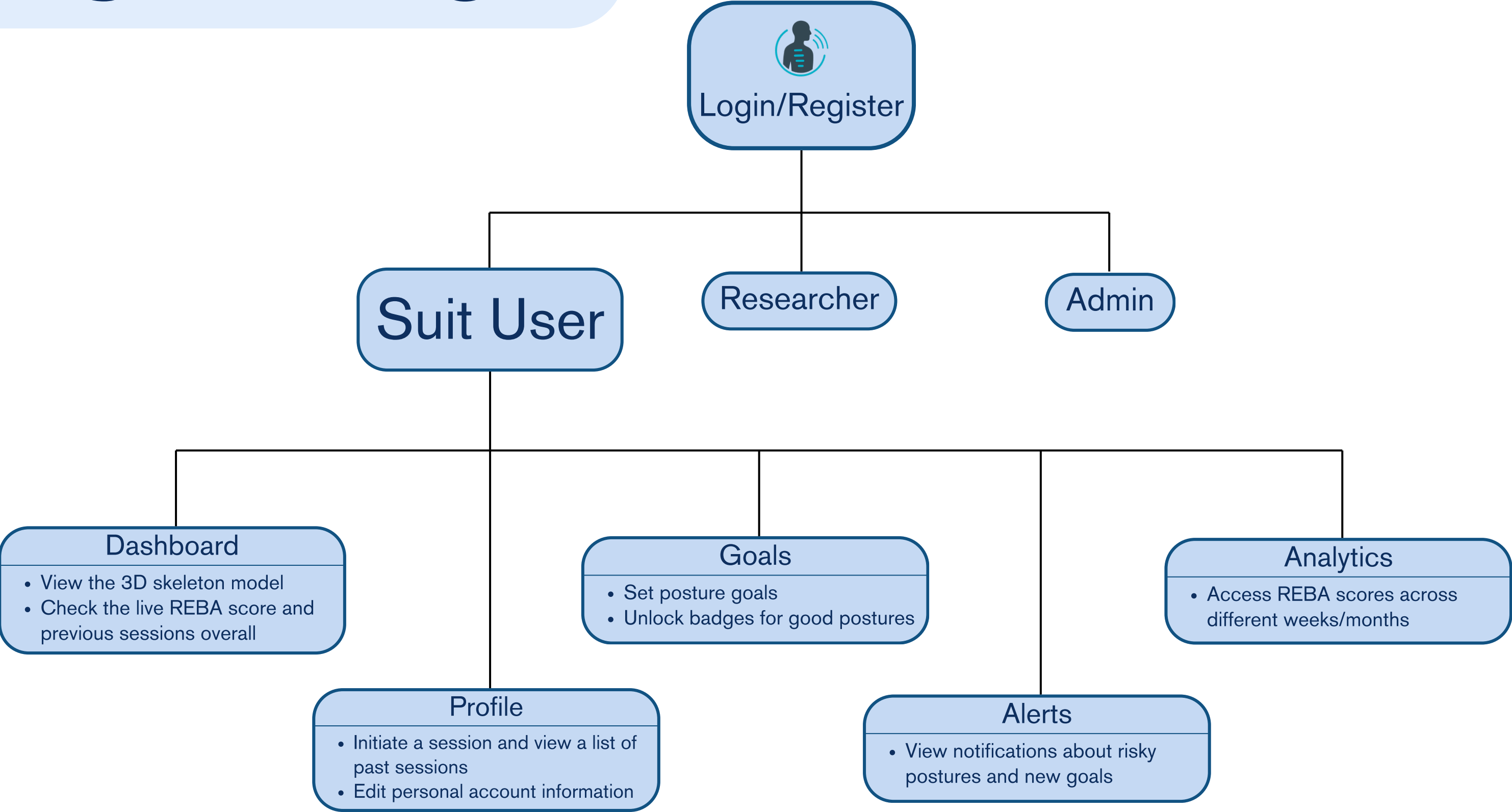
To manage user
accounts

I can control access and
provide support

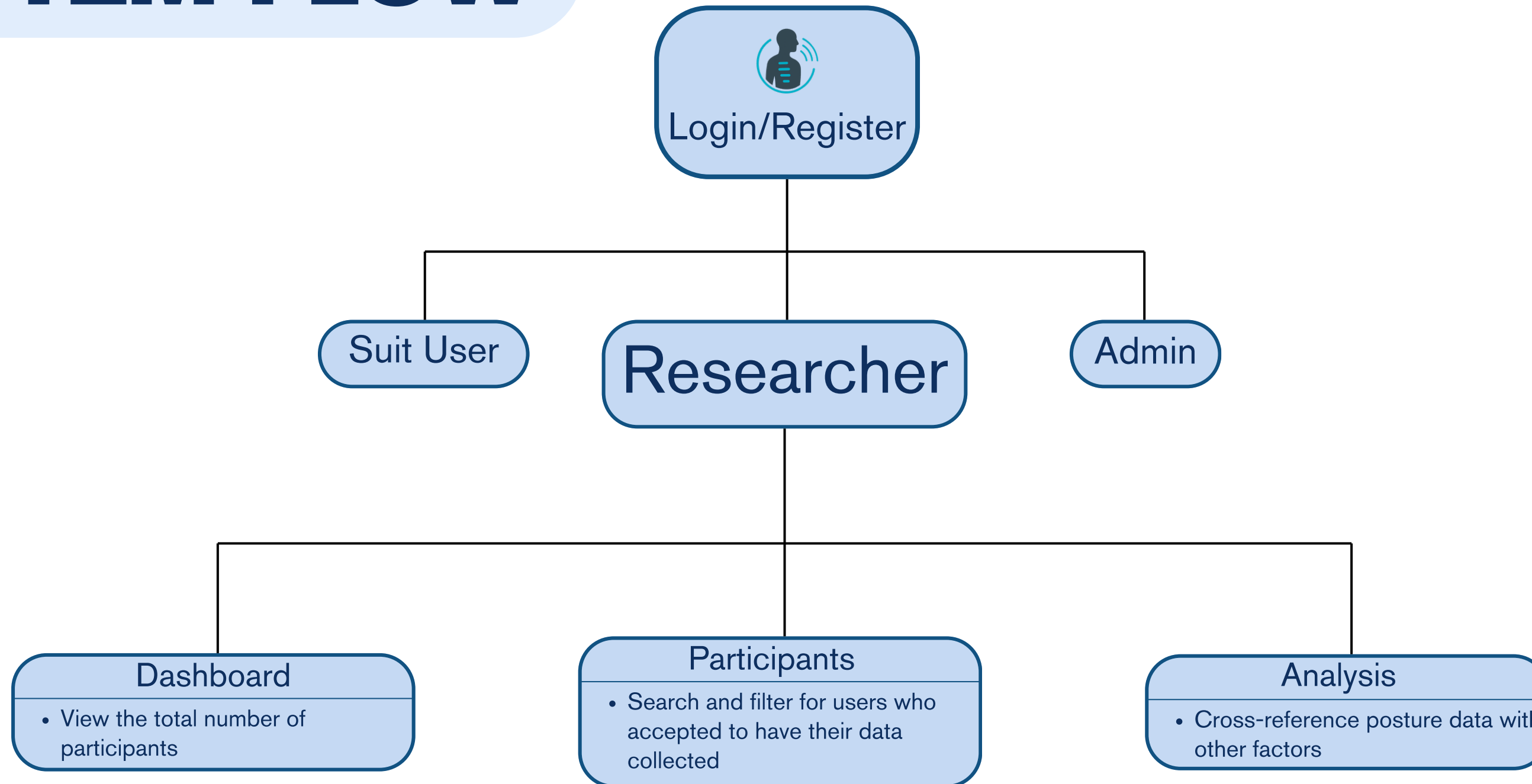
SYSTEM FLOW



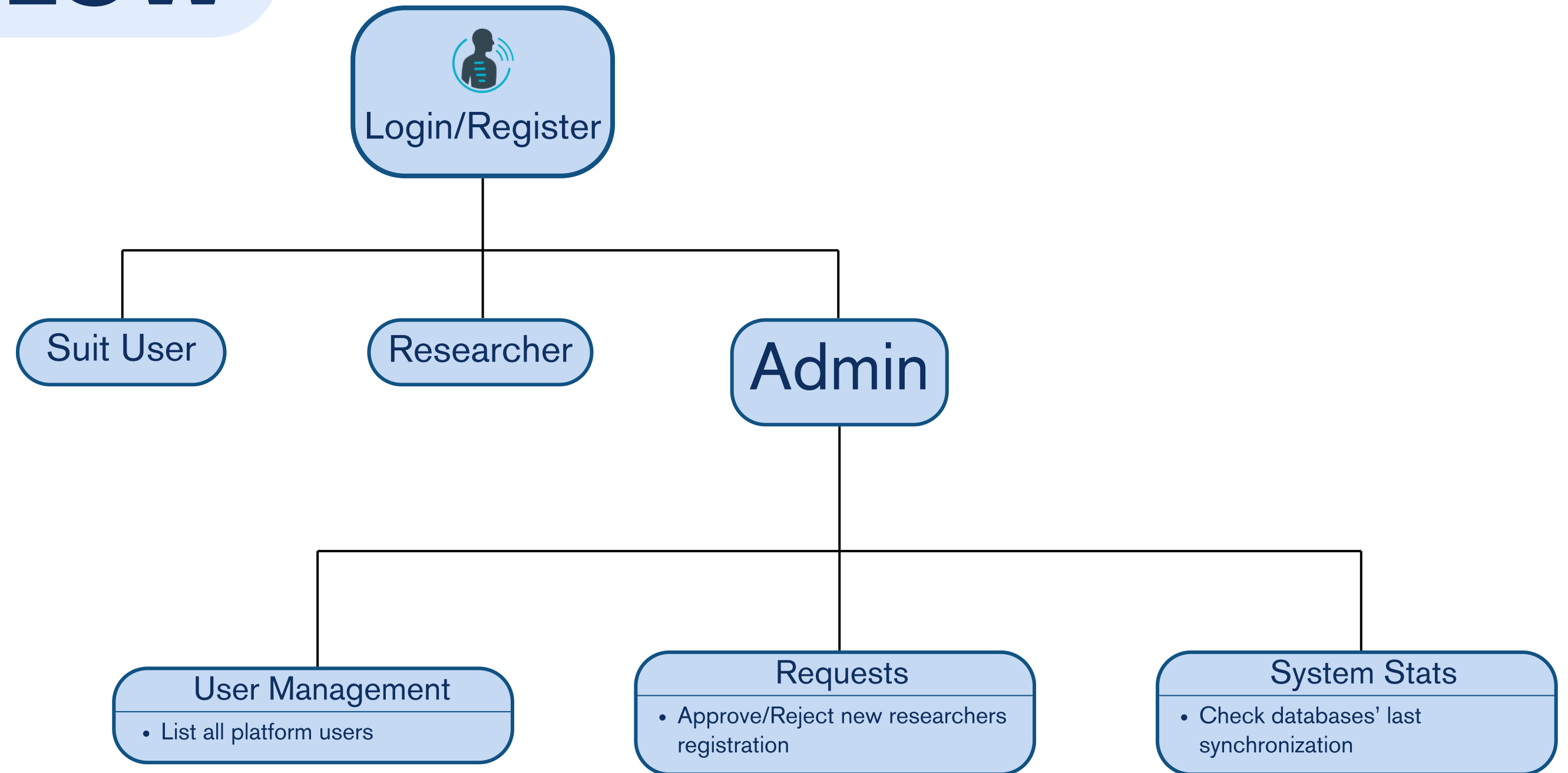
SYSTEM FLOW



SYSTEM FLOW



SYSTEM FLOW



FEEDBACK

from the 2th seminar

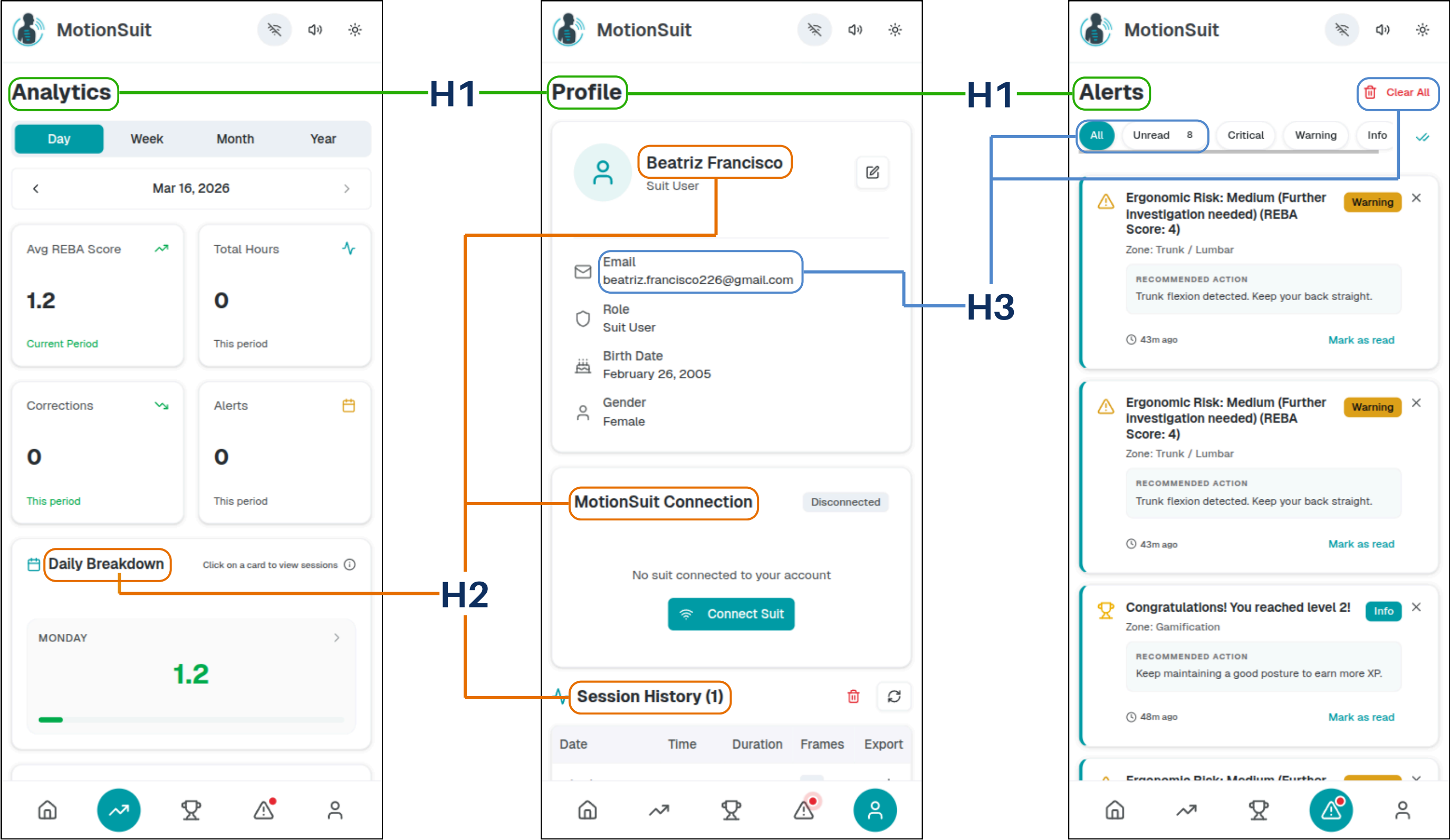
**Change the offline
color from red to
gray.**

**Change the goals
page icon**

**Implement a
hoverboard to improve
user perception.**

**There's a lot of
uniformity in the
admin dashboard.**

VISUAL HIERARCHY



PROGRESS

ANALYSIS



**USER
EXPERIENCE**



PROGRESS

ANALYSIS



1

REBA score

2

Track REBA
scores across
previous sessions

3

Alert system

USER
EXPERIENCE



PROGRESS

ANALYSIS



USER EXPERIENCE



- 1** Accessibility (tabs, text-to-voice, colorblind mode)
- 2** Adaptive dashboard (dark/light mode, customizing charts in the investigator dashboard)
- 3** Gamification

ERROR HANDLING



Welcome back

Email or password incorrect.

[Sign in](#)

beatriz.francisco226@gmail.com

.....

Sign in

Don't have an account? [Sign up](#)

 Personal Details

FIRST NAME *

Beatriz

LAST NAME *

Silva

Last name must be at least 2 characters

EMAIL ADDRESS *

name@example.com

Invalid email address

GENDER *

Female

BIRTH DATE *

02/26/2005



 Security

PASSWORD *

.....

☐ Min 8 chars

☒ Number

☒ Uppercase

☐ Special char

Password must be at least 8 characters

CONFIRM PASSWORD *

.....

Passwords do not match

 Personal Details

FIRST NAME *

Beatriz

LAST NAME *

Francisco

EMAIL ADDRESS *

beatriz.francisco226@gmail.com

GENDER *

Female

BIRTH DATE *

02/26/2005



 Security

PASSWORD *

User123!

☒ Min 8 chars

☒ Number

☒ Uppercase

☒ Special char

CONFIRM PASSWORD *

User123!

 Error

Email already registered



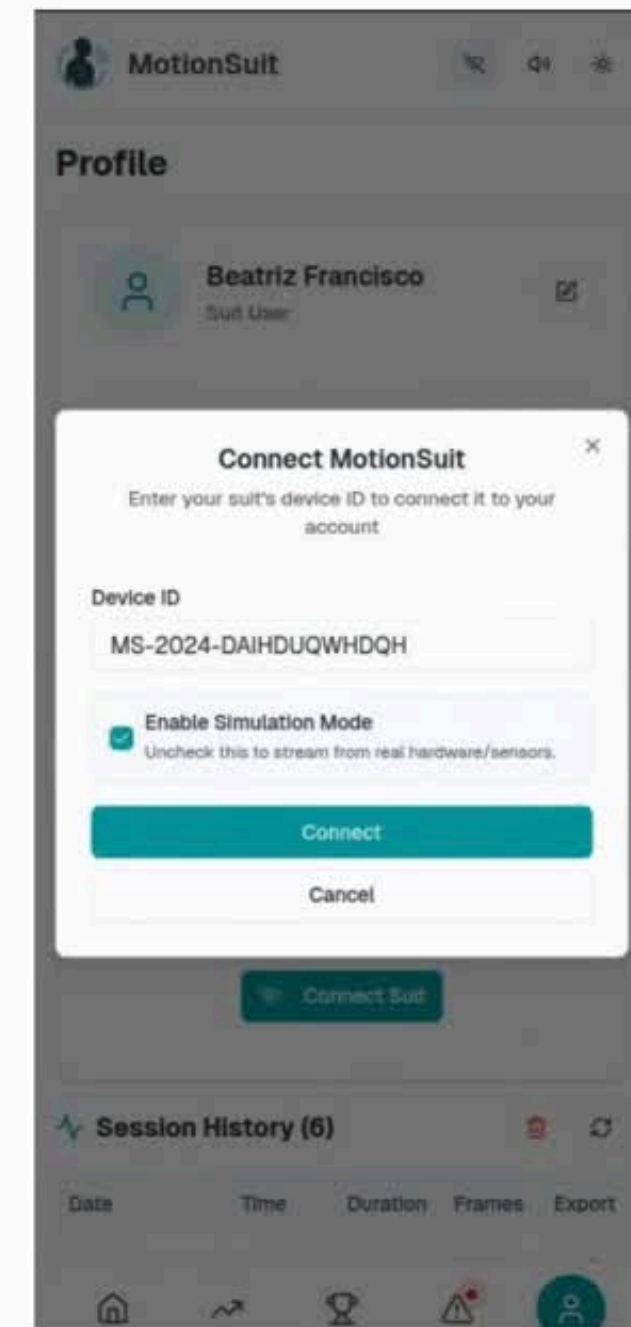
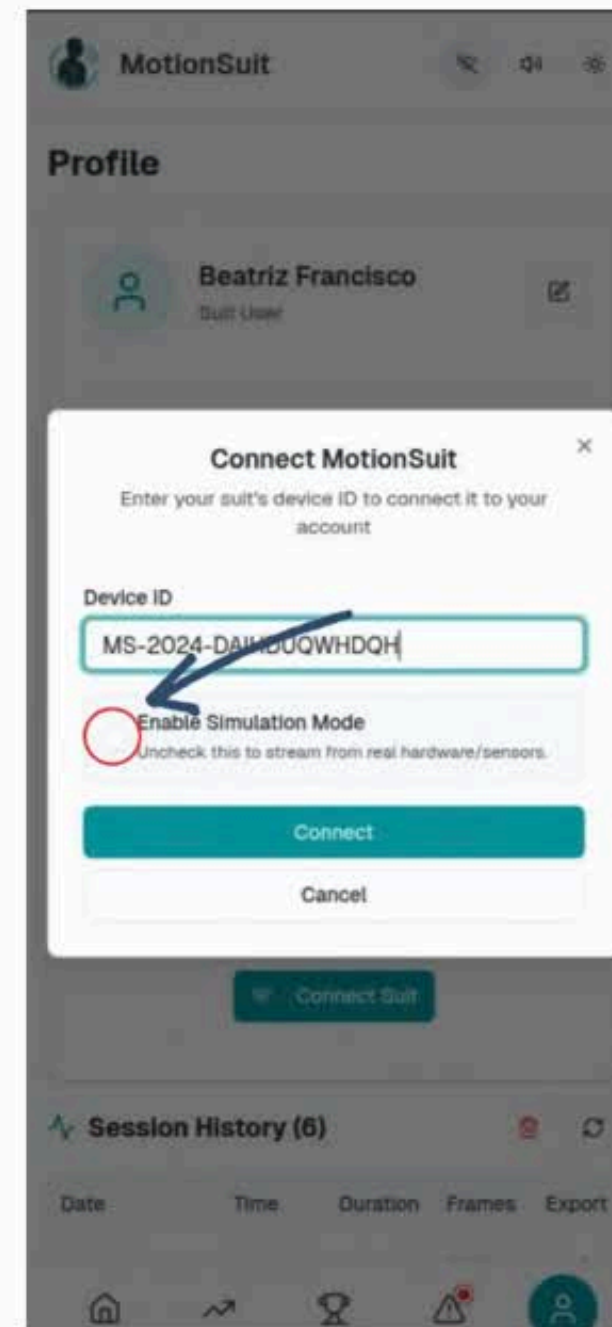
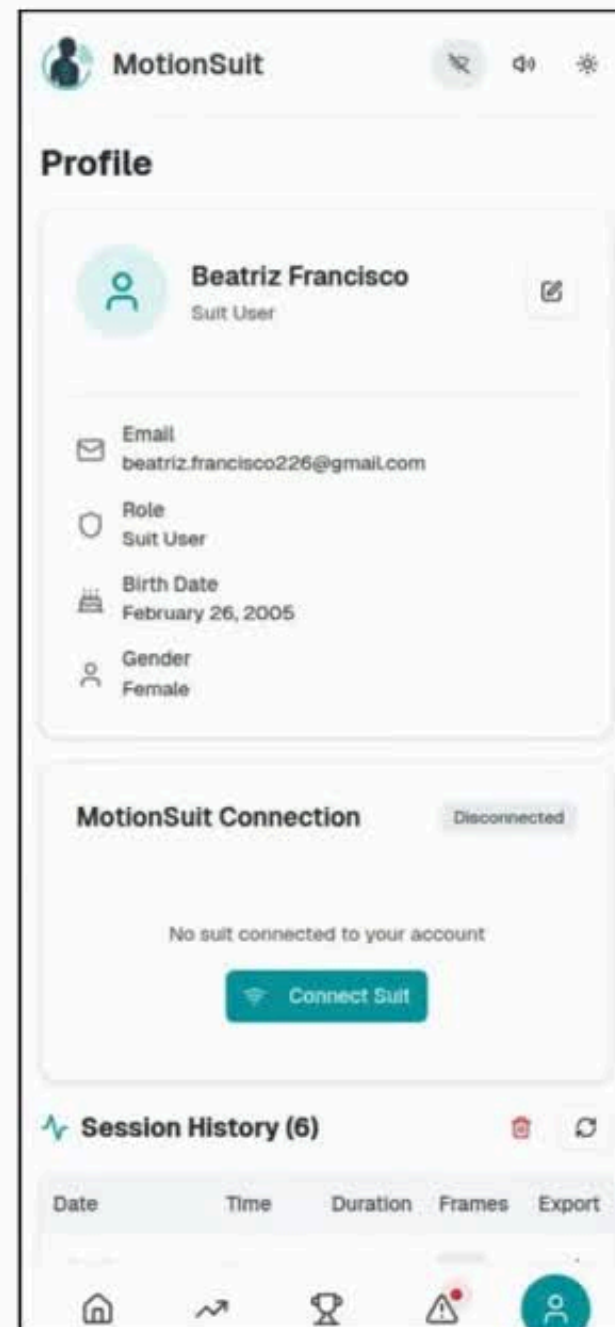
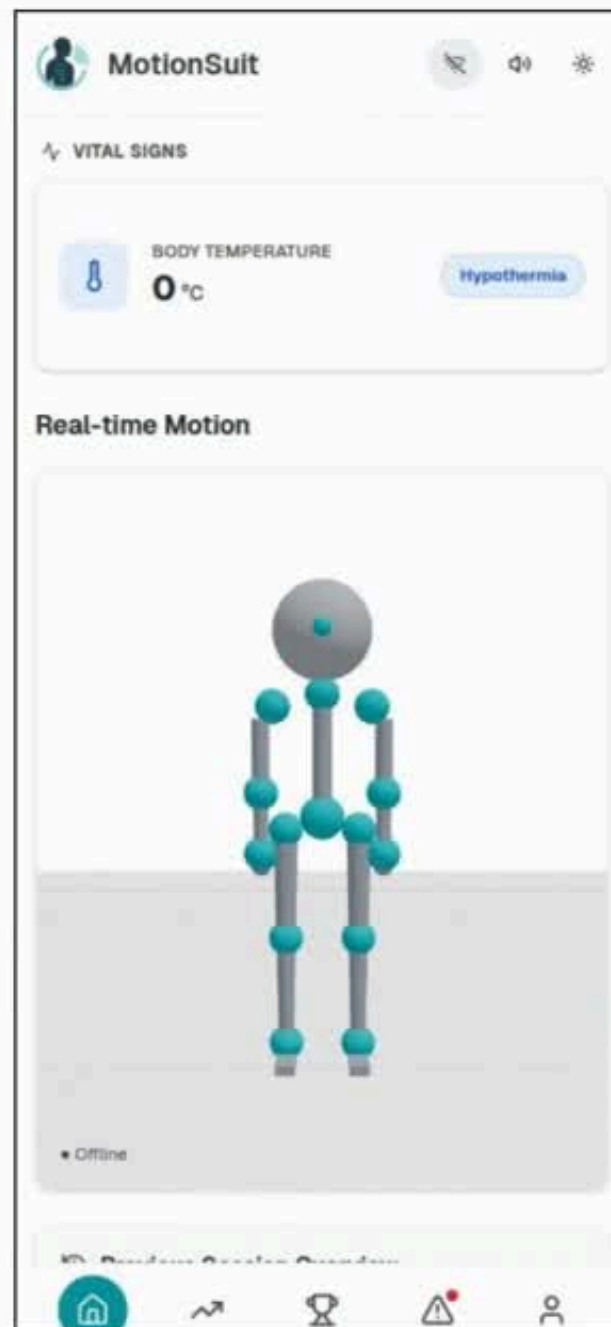
404

Lost in space?

We couldn't find the page you're looking for. It might have been moved, deleted, or perhaps it never existed in the first place.

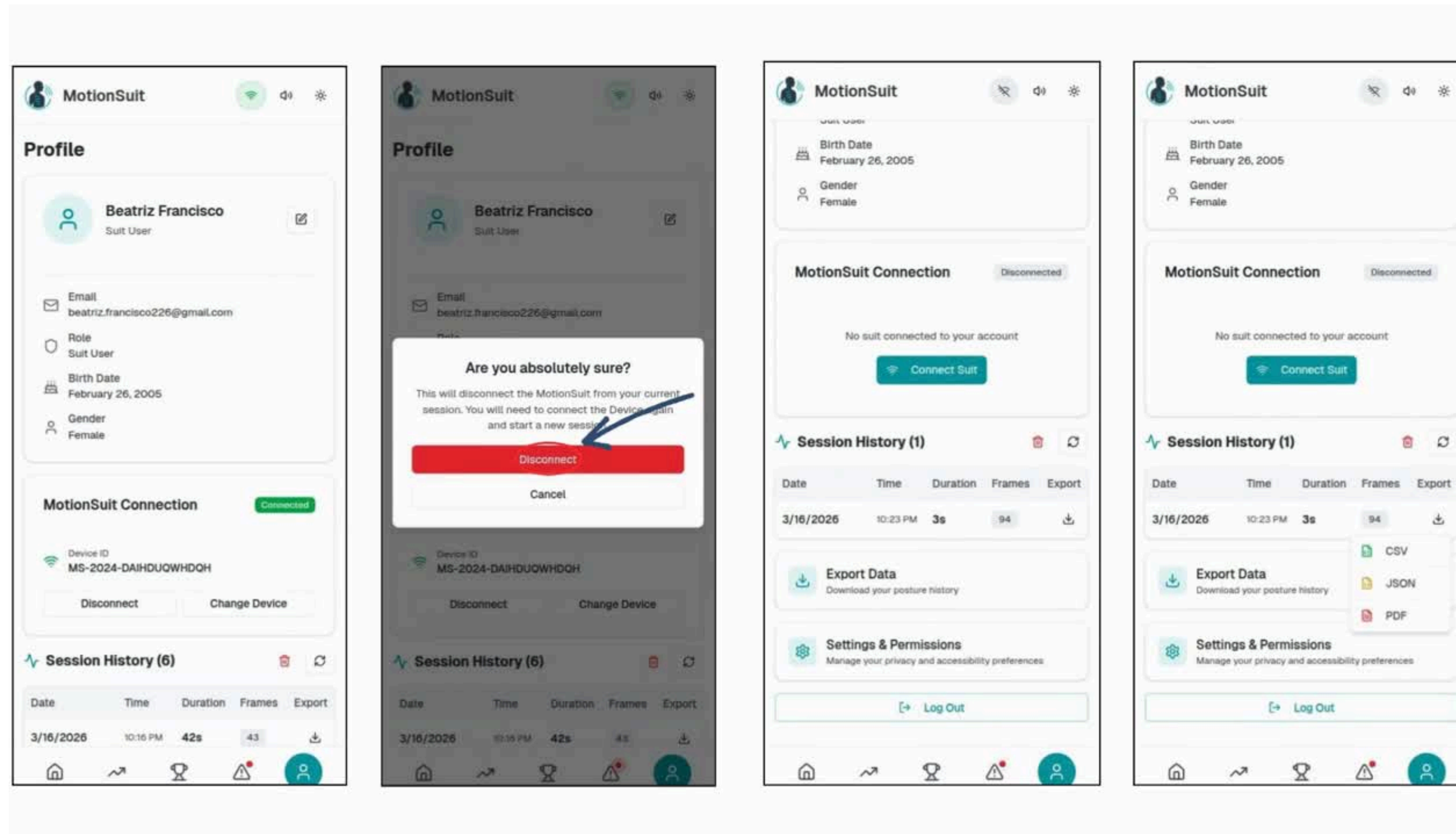
Go to Dashboard

USABILITY

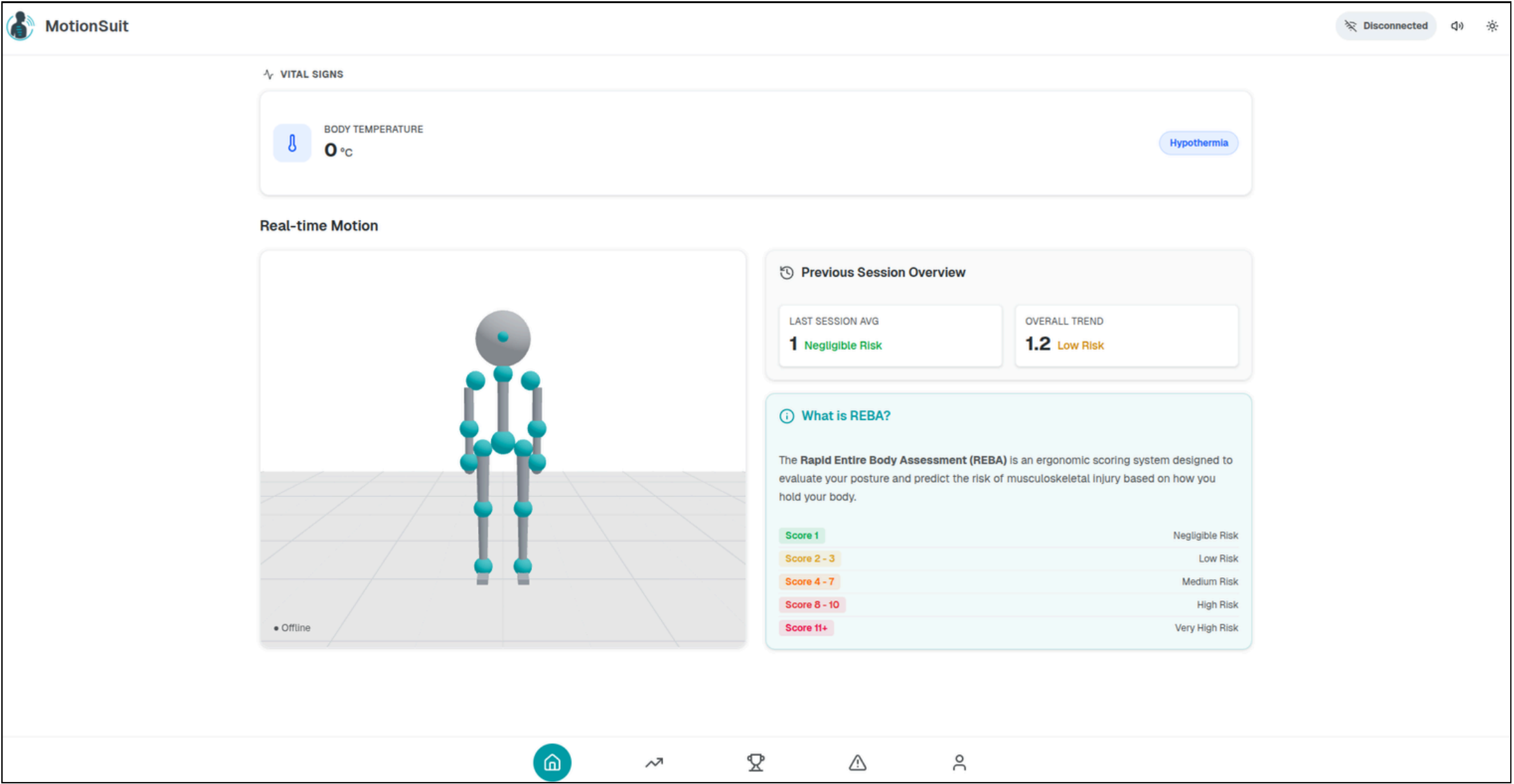
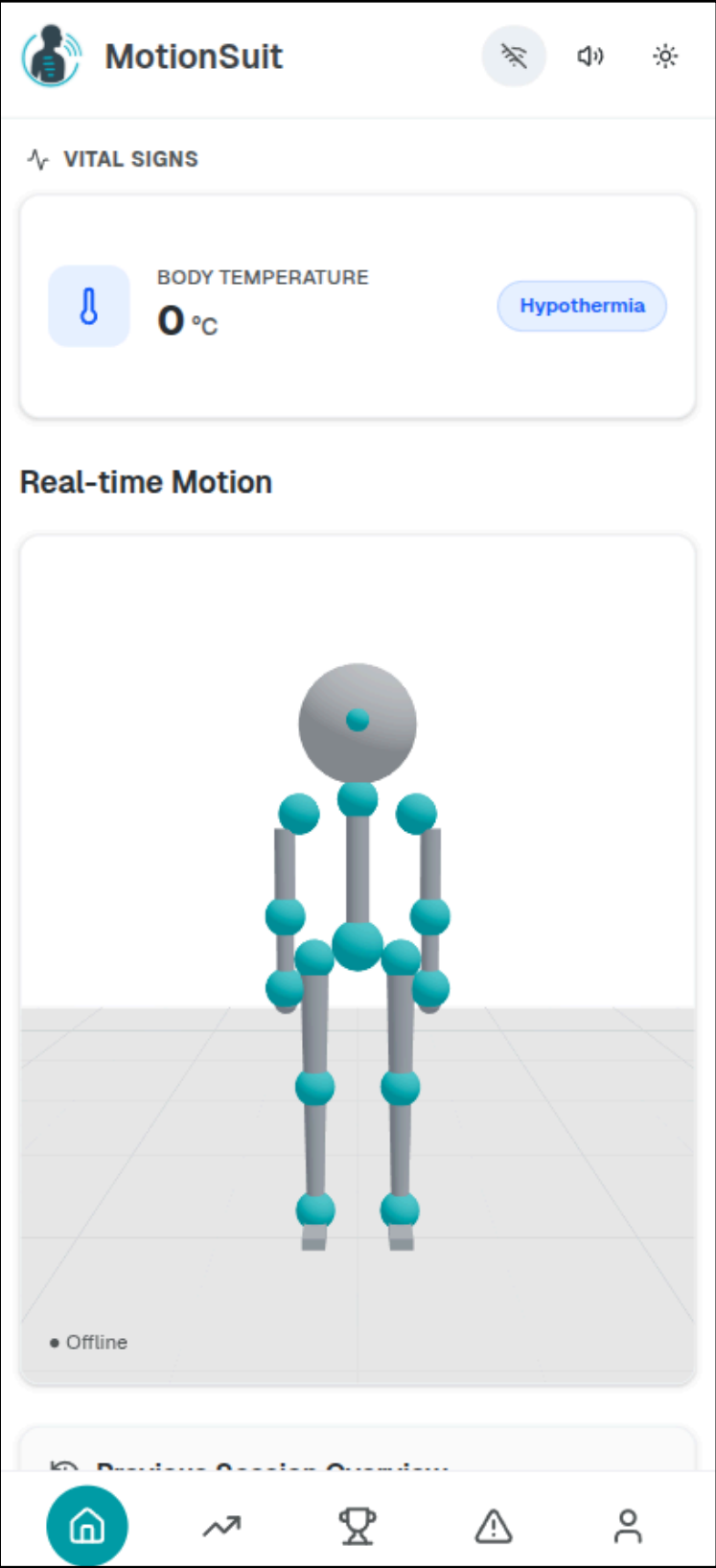


USABILITY

start session, stop session and export data → 8 clicks for 3 actions



VISUAL PERCEPTION



VISUAL PERCEPTION



MotionSuit

Research

Matilde RodriguesLogout

Overview

Participants

Experiments

Analysis

Research Dashboard

Monitor active experiments and participant data

Active Participants

0

Registered users

Total Sessions

0

Data uploads

Data Collected (hrs)

0

Total duration

Active Studies

1

Main Campaign

Active Experiments

General Data Collection

MAIN-2024 • 0 participants

In Progress

Progress0%





MotionSuit Research

Matilde RodriguesLogout

Overview

Participants

Experiments

Analysis

Reports

Research Dashboard

Monitor active experiments and participant data

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0

Registered users

Total Sessions

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Data uploads

Data Collected (hrs)

0

Total duration

Active Studies

1

Main Campaign

Active Experiments

General Data Collection

MAIN-2024 • 0 participants

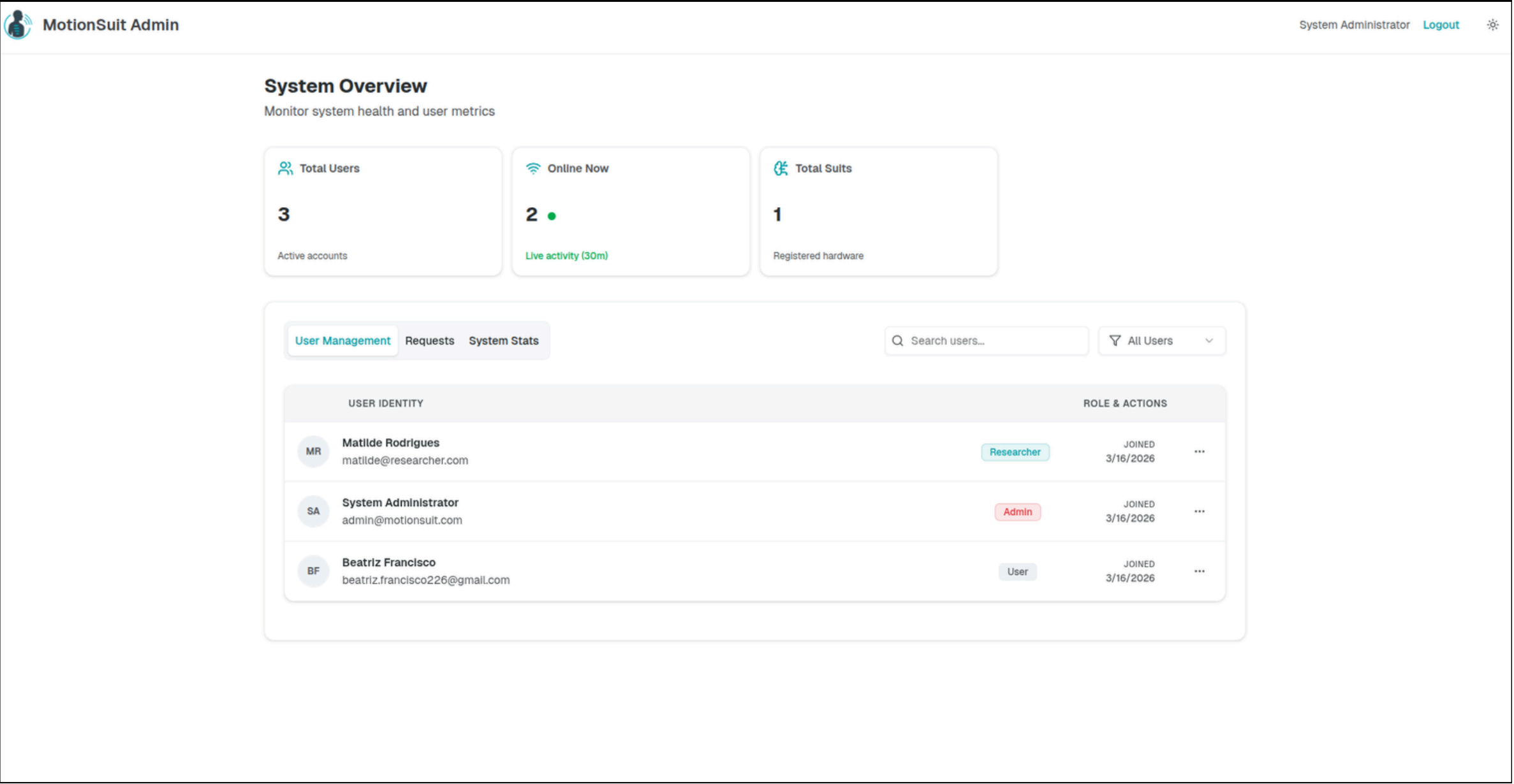
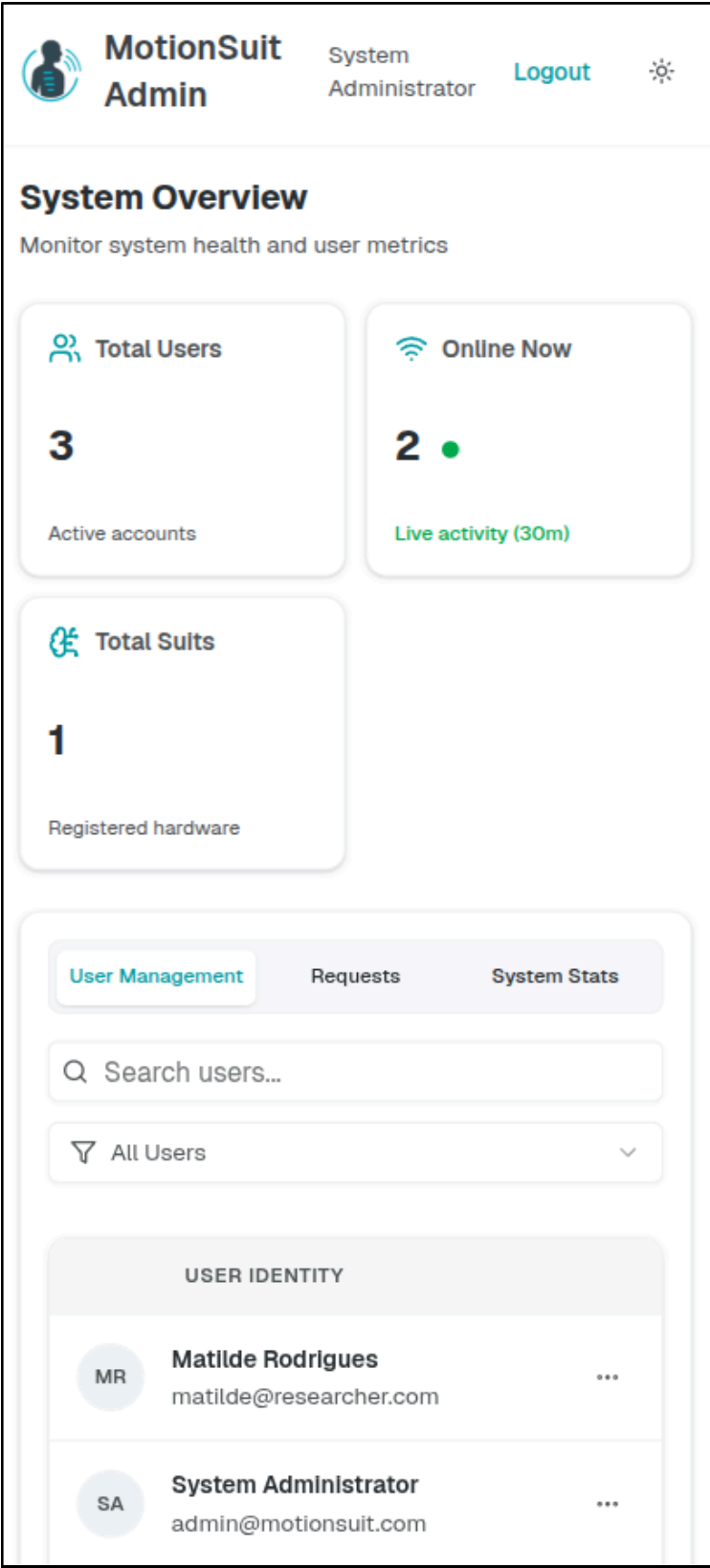
In Progress

Progress0%

Recent Activity

No recent activity recorded.

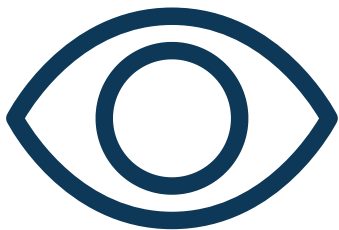
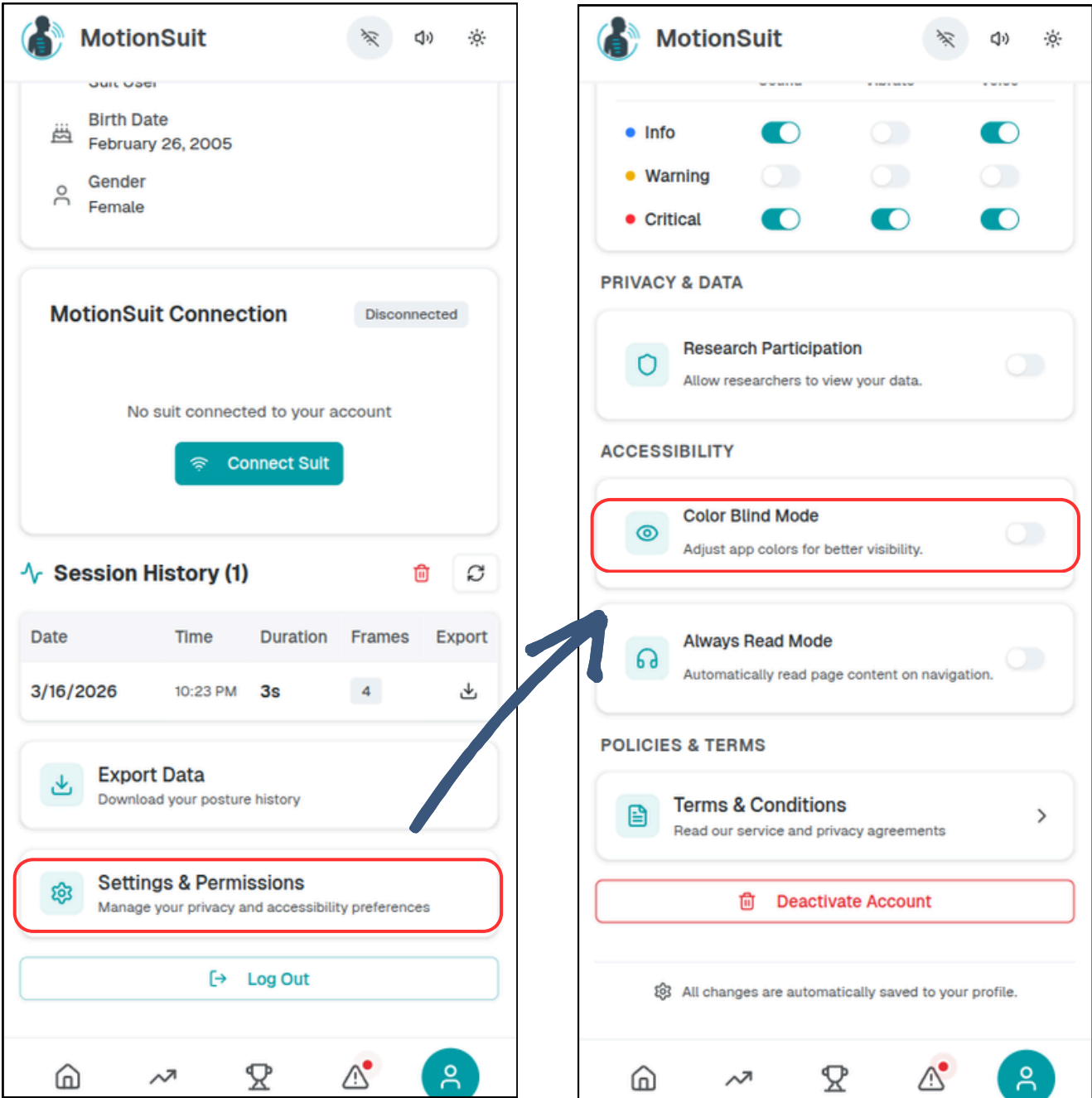
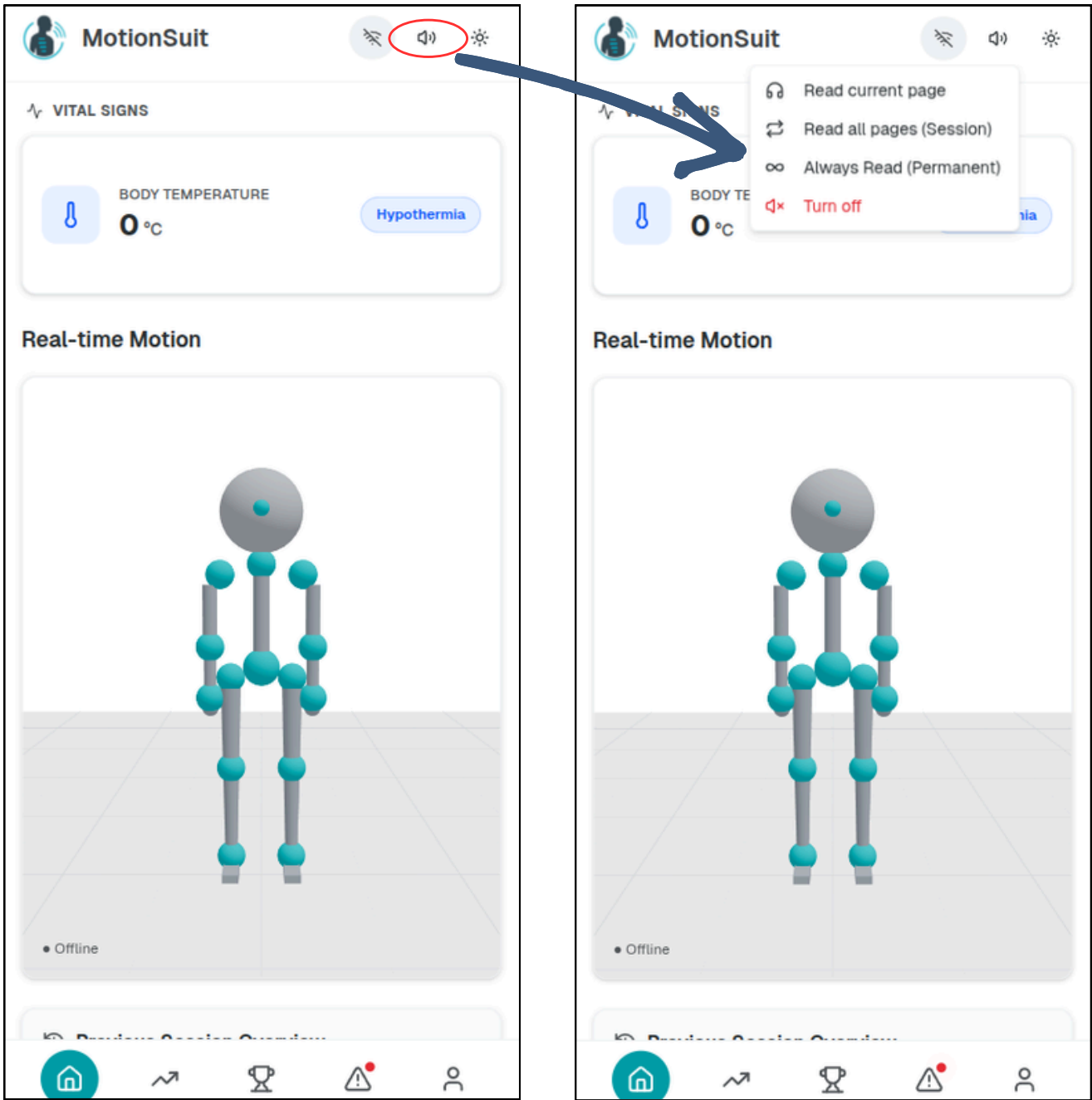
VISUAL PERCEPTION



ACCESSIBILITY



Text-to-Speech



Colorblind Mode

TASKS

Admin

Researcher

Suit User

Suit User

Task 1: Create a new suit user account and set up the biometric profile.

Task 2: Connect a suit to the system and start a real-time monitoring session.

Task 3: Go to the Dashboard, observe the data and finish the session.

Task 4: Access the Session History and export the last session.

Task 5: Export sessions from the last 7 days in CSV format.

Task 6: Enable "Research Participation" option.

TASKS

Suit User

Researcher

Admin

Task 1: Create a new researcher account and check the account status.

Task 2: Review the list of pending researchers and approve the new researcher account request.

Task 3: Change the password for the researcher account.

Task 4: Verify the details of the suit user account and export the session data.

Task 5: Check the system status and export the sessions logs in JSON format.

Admin

TASKS

Suit User

Admin

Researcher

Task 1: Login in the researcher account created check the information on the dashboard.

Task 2: Check the participants list and filter it by age from 18 to 25 years in ascending order.

Task 3: Search for the suit user account created, check the details and export all data in CSV format.

Researcher

SYSTEM USABILITY SCALE

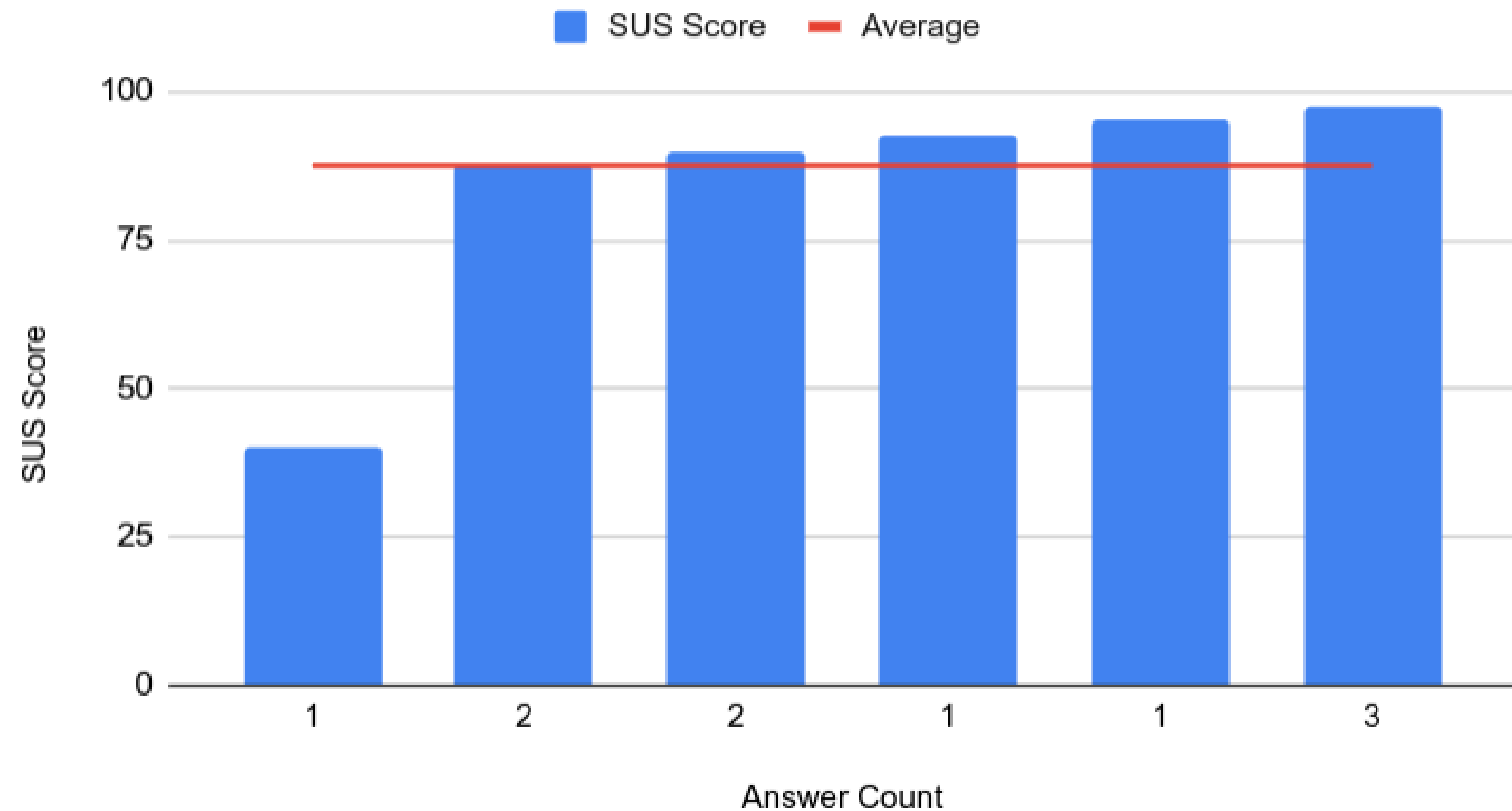


10



87,5

SUS Score of 10 participants



INTERPRETATION

Suit User

Task 2: Connect a suit to the system and start a real-time monitoring session.

Did the user
completed the task
correctly?

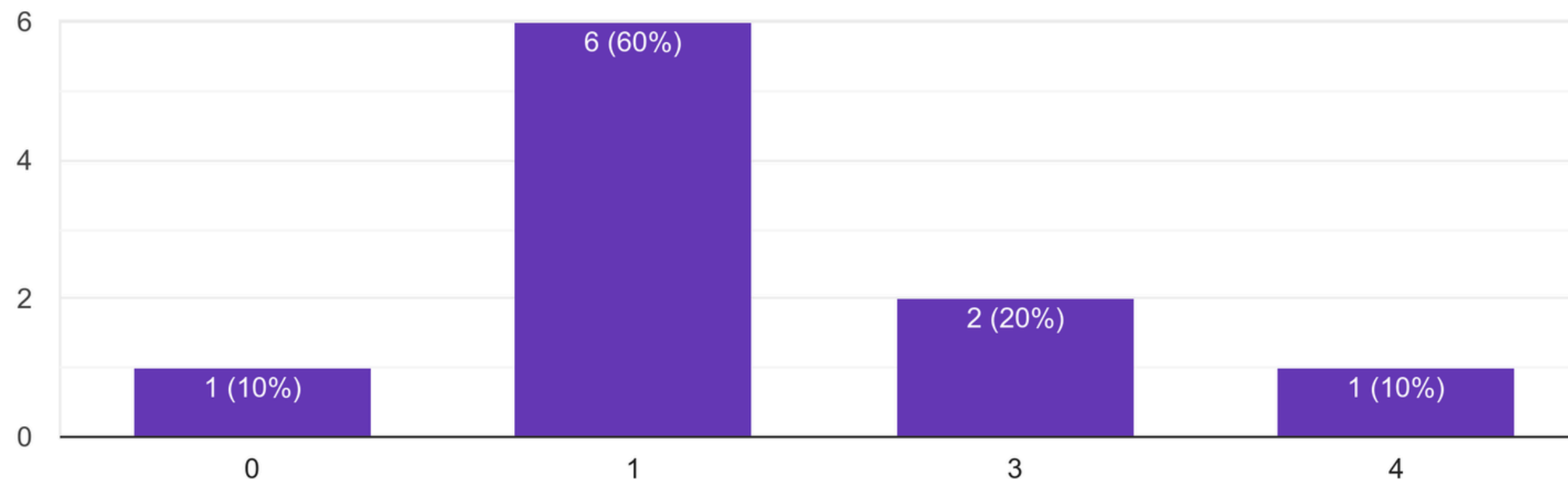
Yes → 100%
No → 0%

Was the user lost?

Yes → 10%
Slightly → 90%
A lot → 0%

Number of errors?

10 respostas



INTERPRETATION

Admin

Task 3: Change the password for the researcher account.

Did the user
completed the task
correctly?

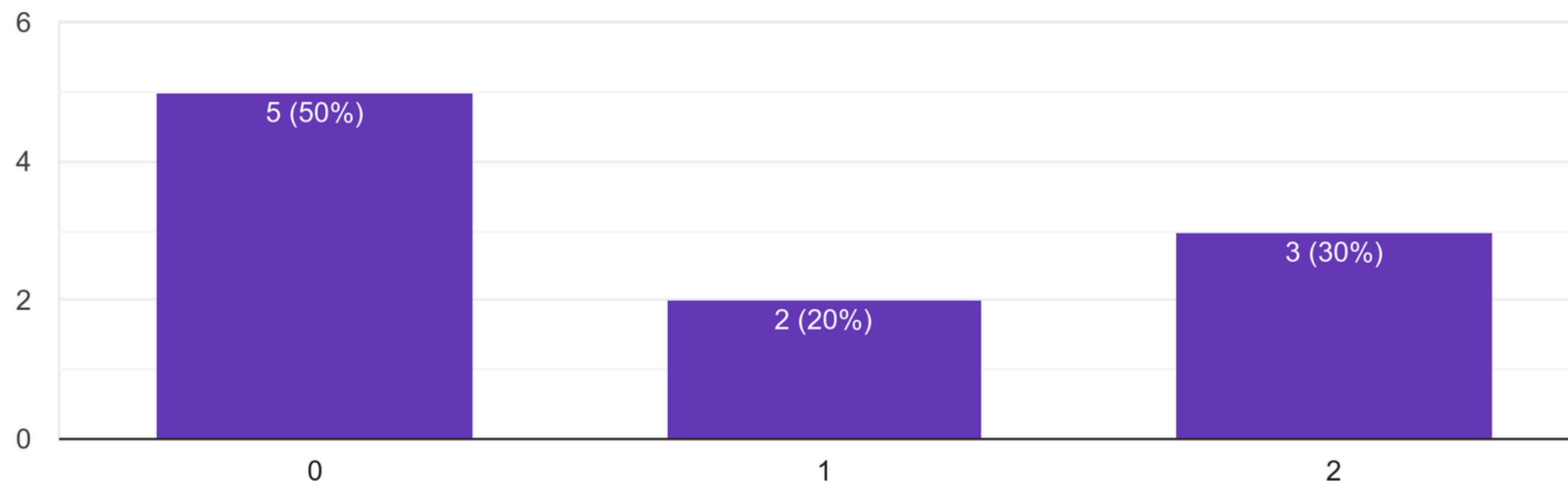
Yes → 100%
No → 0%

Was the user lost?

No → 50%
Slightly → 50%
A lot → 0%

Number of errors?

10 respostas



INTERPRETATION

Researcher

Task 2: Check the participants list and filter it by age from 18 to 25 years in ascending order.

Did the user
completed the task
correctly?

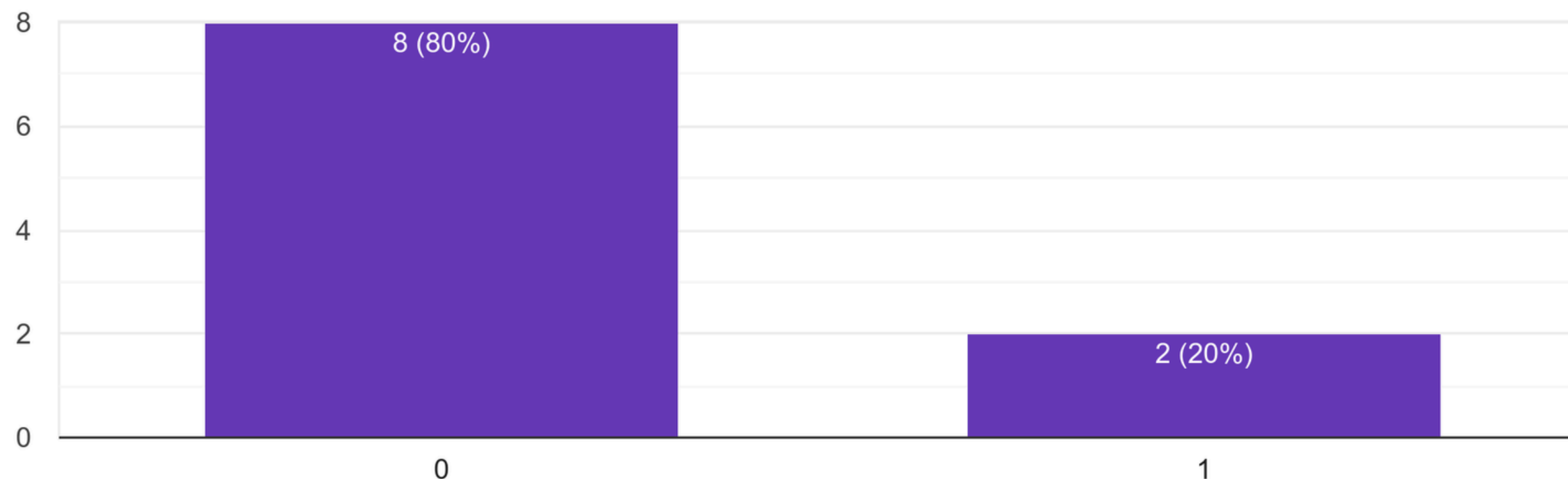
Yes → 100%
No → 0%

Was the user lost?

No → 80%
Slightly → 20%
A lot → 0%

Number of errors?

10 respostas



CALENDAR



March 17

MS1: Digital Accessibility
and Usability

T-1

Done

Receive alerts for incorrect posture

T-4

Done

Customize app alerts

T-11

Done

Analyze collected data



March 24

MS2: Prototype

T-6

In Progress

View real-time progress toward goals

T-5

Done

Set goals to improve posture



<https://pei-motionsuit.github.io/microsite/>

THANK YOU!



<https://pei-motionsuit.github.io/microsite/>